

What is the H Query in Groth16?

The H query relates to the computation of a specific polynomial that arises during the zero-knowledge proof process in Groth16. This polynomial is part of the proving system's setup and is used to ensure correctness and completeness of the proof.

Context of the H Polynomial

In Groth16, the prover needs to demonstrate the validity of a computation encoded as a R1CS and can be represented by three polynomials: $A(x)$, $B(x)$, and $C(x)$, which satisfy the relation:

$$A(x) \cdot B(x) - C(x) = H(x) \cdot Z(x),$$

where:

- $Z(x)$ is a fixed polynomial that encodes the structure of the constraint system.
- $H(x)$ is the quotient polynomial, which ensures that $A(x) \cdot B(x) - C(x)$ vanishes over the roots of $Z(x)$.

The H query in Groth16 consists of group elements representing evaluations of the polynomial $H(x)$ over specific points. These evaluations are precomputed during the trusted setup phase.

How is $H(x)$ Constructed?

Within the “phase1radix2mX” file, $H(x)$ it is computed using Lagrange interpolation and certain transformations. Specifically, $H(x)$ involves terms of the form:

$$x^{i+m} - x^i, \quad \text{for } i = 0, 1, \dots, m-2,$$

where:

- x^i are the powers of the trapdoor τ (a secret scalar used in the setup phase).
- m is the degree of the polynomial or the size of the evaluation domain.

These terms encode the relationship between the different powers of τ and form the basis for $H(x)$.